

TYPES OF RESEARCH BY METHOD/ BY DESIGN

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TYPES OF EDUCATIONAL RESEARCH

TOPIC:

**TYPES OF RESEARCH BY
METHOD/ BY DESIGN**

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Quantitative research and Qualitative research

Quantitative research is the collection and analysis of numerical data to describe, explain, predict, or control phenomena of interest.

Qualitative research is the collection, analysis, and interpretation of comprehensive narrative and visual (i.e., nonnumerical) data to gain insights into a particular phenomenon of interest.



CLASSIFICATION OF RESEARCH B Y METHOD

1. Quantitative Research Approaches

- a) Survey Research
- b) Correlational Research
- c) Casual-Comparative Research
- d) Experimental Research
- e) Single-Subject Research

2. Qualitative Research Approaches

- a) Narrative Research
- b) Ethnographic Research
- c) Case Study Research



Quantitative research approaches

Quantitative research approaches are applied to describe current conditions, investigate relations, and study cause–effect phenomena.

1. Survey research

- is often designed to describe current conditions.

2. Correlational research

- Studies that investigate the relations between two or more variables.

3. Experimental Research and, 4. Causal–Comparative Research

- provide information about cause–effect outcomes.

5. Single-Subject research

- focus on the behavior change an individual
- exhibits as a result of some intervention fall
- under the heading of single-subject research.

1. SURVEY RESEARCH

- It involves collecting numerical data to test hypotheses or answer questions about the current status of the subject of study.
- One common type of survey research involves assessing the preferences, attitudes, practices, concerns, or interests of a group of people. e.g. political party survey etc.
- Survey research data are mainly collected through questionnaires, interviews, and observations.
- **Challenges**
 1. Researchers often ask questions that have not been asked before, they usually have to develop their own measuring instrument for each survey study. Constructing questions requires clarity, consistency, and tact.
 2. Other major challenges are participants' failure to return questionnaires, their willingness to be surveyed over the phone, and their ability to attend scheduled interviews.
 3. If the response rate is low, then valid, trustworthy conclusions cannot be drawn.

2. CORRELATIONAL RESEARCH

- Correlational research involves collecting data to determine (to what degree) a **relation exists between two or more** quantifiable **variables** from a single group of participants.
- A **variable** is a place-holder that can assume any one of a range of values; for example, intelligence, height, and test score are variables.
- The purpose of a correlational study may be to establish relations or use existing relations to make predictions.
- **Correlation** refers to a quantitative measure of the degree of correspondence. The degree to which two variables are related is expressed as a **correlation coefficient**, which is a number between **+1.00** and **-1.00**.
- Two variables that have a correlation coefficient near **0.00** are **not related**.
- Two variables that have a correlation coefficient near **+1.00** or **-1.00** are **highly correlated**.
- A number near **+1.00** indicates a **positive correlation**: As one variable increases, the other variable also increases.
- A number near **-1.00** indicates a **negative correlation**: As one variable increases, the other variable decreases.

Note:

It is very important to note that the results of correlational studies do not suggest cause– effect relations among variables.

3. CAUSAL–COMPARATIVE RESEARCH

- **Causal–Comparative research** attempts to establish cause–effect relations among groups.
- to determine the **cause, or reason**, for existing differences in the behavior or status of groups of individuals.
- A weakness of causal–comparative studies is that, the researcher has no control over it because the cause under study has already occurred .
- Although causal–comparative research produces limited cause–effect information, it is an important form of educational research. True cause–effect relations can be determined only through experimental research in which the researcher maintains control of an independent variable but in many cases, an experimental study is inappropriate or unethical.
- The causal–comparative approach is chosen precisely because the grouping variable either cannot be manipulated (e.g., as with gender, height, or year in school) or should not be manipulated.
- A causal–comparative study, which approximates cause–effect results without harming the participants, is the only reasonable approach.
- However, causal–comparative research does not produce true experimental research outcomes.

Example

- ***attendance*** affect ***maturity*** at the end of the first grade